

Zhongxin Hu

PhD Student in Eindhoven University of Technology

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Research Profile

My research focuses on sustainable operations management. I study circular business models, such as remanufacturing and refurbishment, that enable firms to make decisions that are economically, environmentally, and (hopefully) socially beneficial.

My broader research interests include pricing and equilibrium analysis in queueing systems, behavioral operations, and sequential games.

Educational Background

Eindhoven University of Technology (TU/e)

Eindhoven, the Netherlands

Ph.D. Candidate in Operations Management

July 2023 - Ongoing

- Supervisors: Zümbül Atan and Christina Imdahl
- Group: Operations, Planning, Accounting & Control (OPAC)

Beijing University of Posts and Telecommunications (BUPT)

Beijing, China

M.Sc. in Mathematics

Sep. 2020 - Jun. 2023

- Supervisor: Yongjiang Guo
- Honor: Outstanding Graduate Student Award
- Thesis: *Information Disclosure and Reference Dependence in the Cooperative Newsvendor Problem*

Nanjing University of Posts and Telecommunications

Nanjing, China

B.Sc. in Applied Statistics

Sep. 2015 - Jun. 2019

- Honor: Merit Undergraduate Student Award

Working Papers

• Profit Meets Purpose: Refurbish-to-Lease Model for Medical Essentials

Zhongxin Hu, Christina Imdahl, Yeqing Zhou (RSM), Zümbül Atan.

Status: Under Review.

Conference presentations:

2026 OPAC Conference (*Best Presentation Award*), Heeze, NL; Circularity and Sustainability Workshop (Poster), Utrecht, NL.

2025 INFORMS Annual Meeting, Atlanta, USA; Beta Symposium, Soesterberg, NL.

• Strategic Selection of Remanufacturing Business Models: A Consumer Perception Perspective

Zhongxin Hu, Christina Imdahl, Zümbül Atan.

Status: Under Second-round Review.

Conference presentations:

2026 OPAC Conference, Heeze, NL.

2025 International Symposium on Inventories, Budapest, HU.

Publications

- [1] Cao, J., Guo, Y., **Hu, Z.** *The Effect of Loss Preference on Queueing with Information Disclosure Policy.* Methodology and Computing in Applied Probability, 25: 75, 2023. DOI: 10.1007/s11009-023-10047-w
- [2] **Hu, Z.**, Guo, Y. *Cooperating Strategy of Newsvendor with Advertisers Under Partial Quality Disclosure.* Journal of Systems Science and Mathematical Sciences, 43(3): 667–682, 2023. DOI: 10.12341/jssms22073

Project Experiences

LINear to CIRCular Transition (LINCIT)

July 2023 - June 2026

- *Role: Researcher*
- Studied operational decisions in circular business models, e.g., remanufacturing and refurbishment.
- Presented annual progress reports to consortium partners at the yearly consortium meeting.

Post-Graduate Innovation and Entrepreneurship Project

Apr. 2022 - Nov. 2022

- *Role: Project Leader*
- Integrated sequential game-theoretic models with prospect theory and contract theory to optimize computational resource allocation in blockchain.
- Conducted comprehensive literature review of 70+ papers on cooperation contracts, resource allocation in blockchain, supply chain and other application scenarios.

China Contemporary Undergraduate Mathematical Contest in Modeling

Sep. 2017

- Developed joint pricing and allocation model for crowd sourced task platforms and implemented numerical experiments to develop optimal allocation policies.
- Award: Provincial First Prize, Jiangsu Province.

Teaching Experiences

Teaching Assistant

- **BSc Course:**
Fundamentals of Operations Management (TU/e, Q3 2024/25 & 2025/26);
Probability Theory and Stochastic Processes (BUPT, Spring 2021);
Advanced Mathematics (BUPT, Fall 2020).
- **MSc Course:**
Reinforcement Learning: Theory and Applications (BUPT, Summer 2021)

Supervising

- Supervisor of Bachelor End Projects (TU/e, 2024/25 – 2026/27)

Other Activities

Volunteer Mentor , Lumiere Research Inclusion Foundation, Lumiere Education	2026
Session Chair , INFORMS Annual Meeting, Atlanta, USA	2025
Reviewer , <i>Journal of Systems Science and Mathematical Sciences</i>	2023
Fellow , SheShapes Academia 2022, Global Shapers Community Beijing Hub	2022

Personal Interests

I am passionate about exploring flea market and practical upcycling. I also enjoy spending time off-grid with tent camping and photography.